TIME APPLIANCES
(TAP)

Raspberry Pi Compute Module based Open Time Server Cluster (OTS-C) for Resilient Timing at the Edge



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Chief Technologist at ENSCO

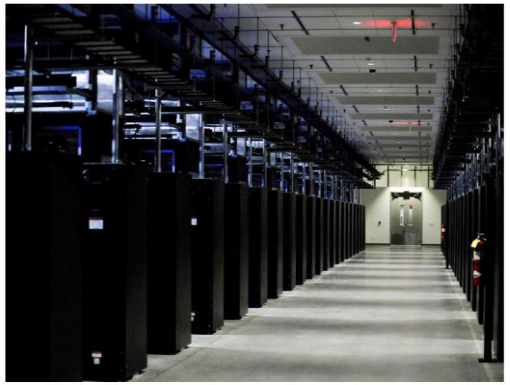
Need for Timing Resiliency at Data Centers

Data Centers need Precise and Synchronized Timing

- **Maintaining Security**
- **Optimizing time-dependent applications involving AI, 5G, and financial trading**
- **Ensuring Data Integrity**



https://www.datacentermap.com/datacenters/



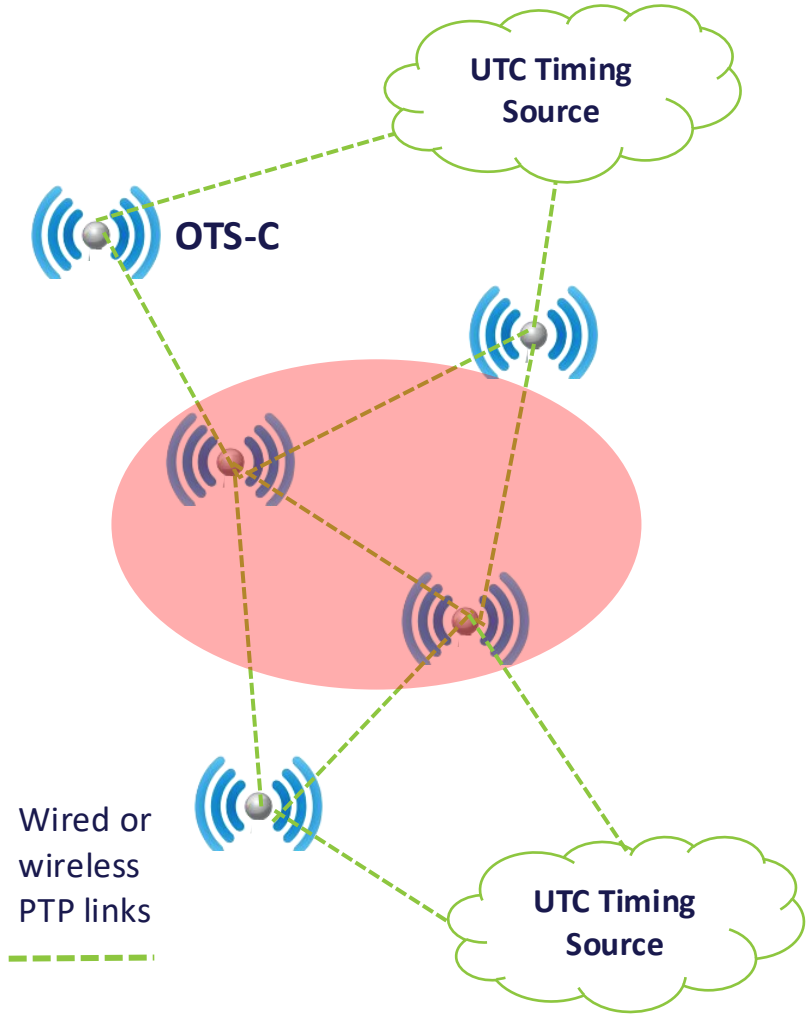
Country	Data Centers
USA	3897
Germany	464
Indonesia	150
Brazil	188
The Netherlands	193
Australia	261
Russia	180
France	284
India	268
Canada	280
United Kingdom	469
Sweden	103

Open Compute Project Time Appliances Project

- Provides high accuracy, reliable, scalable distribution and synchronization of time
- Provides a platform to bring together the community, discuss, standardize and share technologies and solutions across industries relevant to an open-source community
- Targets datacenter applications and datacenter network infrastructure
- Open to the public and welcome all those who would like to be involved
- Open Time Server – Cluster (OTS-C) builds on the **Open, Scalable and Validated** reference architecture of Open Time Servers (OTS) but extends it to support **cluster configurations** for improved resiliency

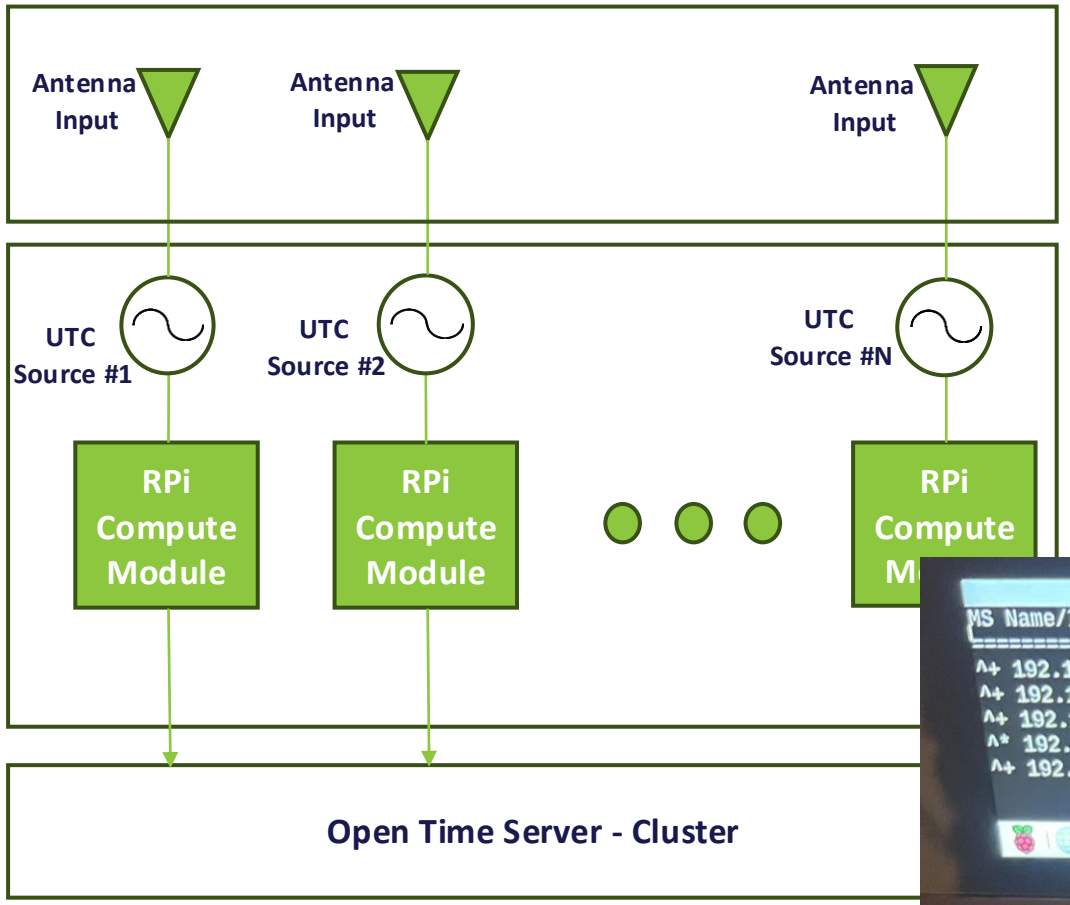
Open Time Server – Cluster (OTS-C)

- OTS-C approach is part of the Open Compute Project seeking resilient timing solutions with respect to the Time Appliances Project (TAP) at the edge
- Open Time Server Cluster (OTS-C) Specification is a timing reference architecture that builds on top of the Open, Scalable and Validated reference architecture of Open Time Servers (OTS) with respect to cluster configurations
- The OTS-C timing reference architecture is built on a PTP network configuration that integrates with open time server deployments for Data Centers (DCs) and edge environments

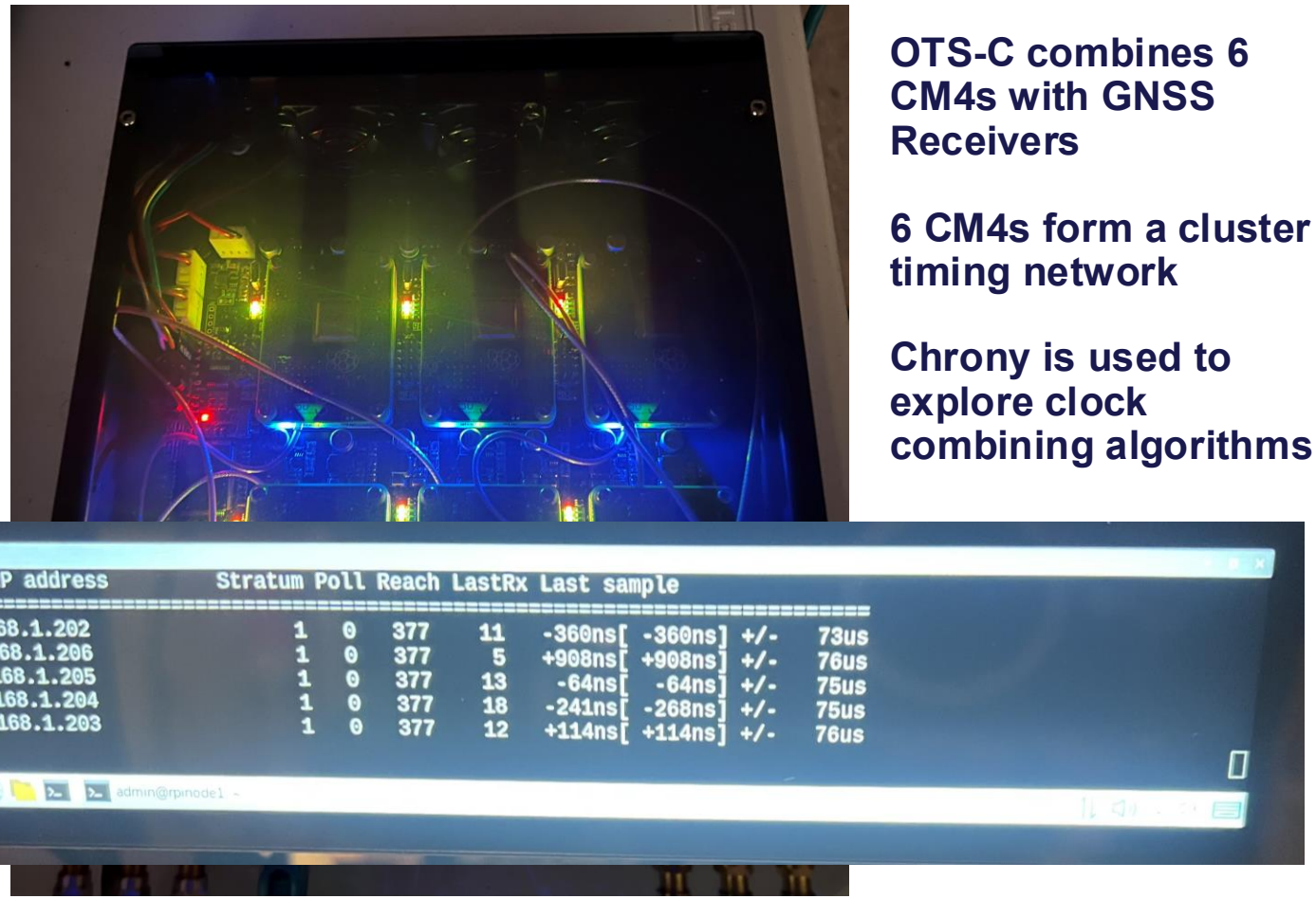


OTS-C Edge Development System with Chrony

OTS-C Block Diagram

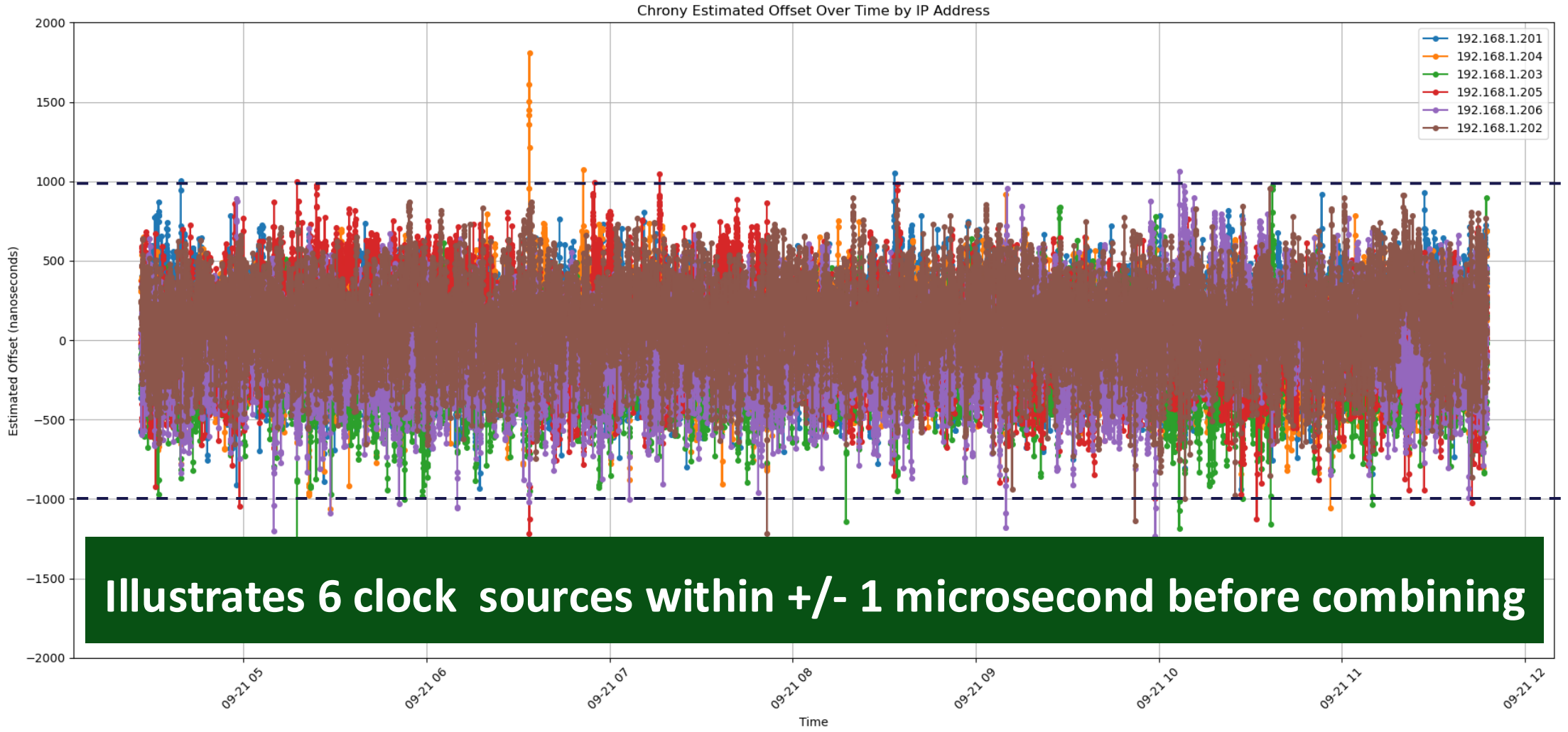


OTS-C Development System



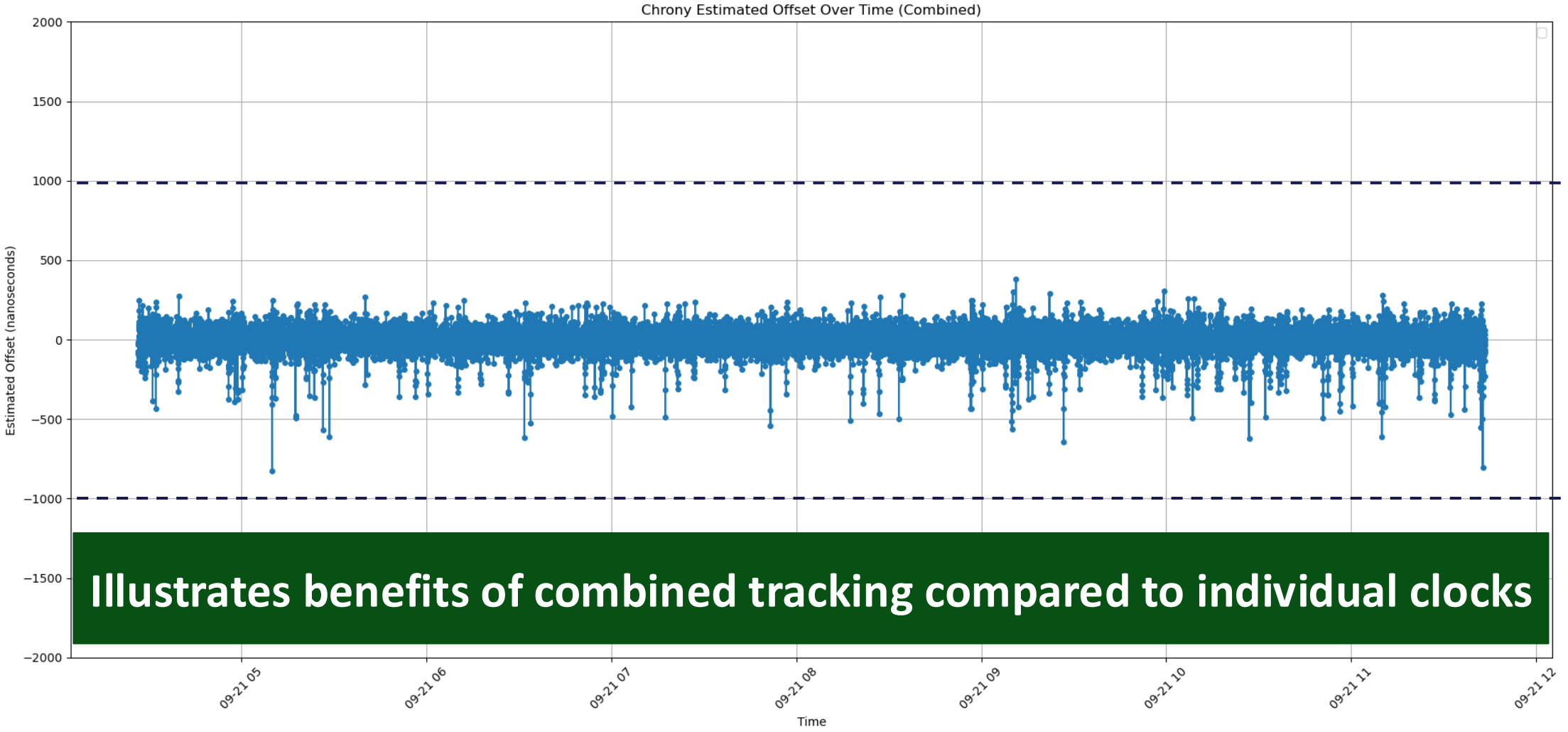
- OTS-C combines 6 CM4s with GNSS Receivers
- 6 CM4s form a cluster timing network
- Chrony is used to explore clock combining algorithms

OTS-C Edge Individual Clock Tracking with Chrony



Illustrates 6 clock sources within +/- 1 microsecond before combining

OTS-C Edge Combined Tracking with Chrony



Illustrates benefits of combined tracking compared to individual clocks

Call to Action for OTS-C Edge Specifications

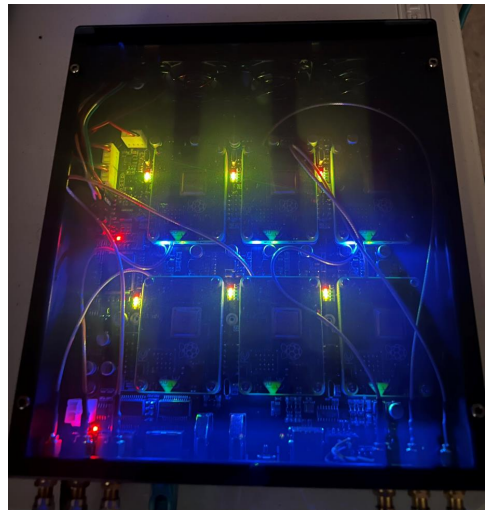
Data Centers - Database | Time Appliances Project - Open

opencompute.org/wiki/Time_Appliances_Project

Workstreams

	Project	Objective	Lead	Mailing List	Status
#1	Time Card	Development of the Time Card	Ahmad Byagowi, Ph.D.		Submitted
#2	Open Time Server	Development of an open time server for DC and Edge systems	Dotan Levi Oleg Obleukhov		Submitted
#3	Data Center PTP Profile	Development of a PTP Profile tailored for data center applications	Michel Ouellette		Submitted
#4	Precision Time APIs	Time APIs to disseminate the time error (error bound) and bring accurate time to the user space	Georgi Chalakov	OCP-TAP-APIs	Submitted
#5	Oscillators	Classification and measuring of oscillators	Gary Giust, Ph.D.		Submitted
#6	PTP Servos	Design and Implement Advanced PTP Servos	Alon Regev		Work in progress
#7	Instrumentation and Measurement	Open source instrumentation and measurement/testing tools for PTP	Anand Ram Julian St. James		Work in progress
#8	Precision Time Synchronization over Wireless	Open source Implementations of Precision Time Synchronization over Wireless	Nobuyasu Shiga, Ph.D. Julian St. James		Work in progress
#9	Precision Time Measurement Readiness Status	List of Hardware and Software with Precision Time Measurement Support	Kevin Stanton, Ph.D. Julian St. James		Work in progress
#10	Time Synchronization Industry-Academia workstream (TSIAW)	Academic and Industry Engagement for Advancement of Precision Time Applications	Dan Biederman Hesham Albakoury		Work in progress
#11	Open Atomic Ethernet (OAE)	Open Atomic Ethernet (OAE)	Paul Borrill, Ph.D. Sahas Munamala	OCP OAE	Work in progress
#12	Open Time Server Cluster (OTS-C)	Specification for a cluster built on top of OTS	Wil Myrick, Ph.D.	OTS-C	Initiated
#13	Unified Intelligent Infrastructure (UII)	Driving collaboration, innovation and coherence across hybrid multi-cloud infrastructure, AI systems, distributed architectures, time-aware computing, and emerging domains such as quantum and beyond.	Ankur Sharma	UII	Initiated

OTS-C Development System



OCP-TAP Wiki Captures Initiation of OTS-C Project to explore Open Time Server Cluster architectures

Thank You!

